

Talha Mehboob

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EDUCATION

University of Massachusetts Amherst

Ph.D. in Computer Engineering

Amherst, MA

Sep 2021 – Sep 2026 (Expected)

- Advisors: Prof. David Irwin & Prof. Michael Zink

National University of Sciences & Technology (NUST)

B.S. in Electrical Engineering

Islamabad, Pakistan

Sep 2017 – Jun 2021

- Rector's Gold Medal — best Final Year Project & Thesis across the entire graduating class

RESEARCH EXPERIENCE

Pacific Northwest National Laboratory (PNNL)

Research Scientist Intern

Richland, WA

May 2024 – Aug 2024

- Designed and implemented PowerTrip, a large-scale distributed ML training infrastructure system for foundation model training across geo-distributed GPU clusters — published at ACM SoCC '25 with PNNL researchers Dr. L. Guo and Dr. N. Tallent.
- Built a GPU cluster profiling module for real-time characterization of compute capacity and network bandwidth/latency across distributed sites — enabling data-driven scheduling decisions across heterogeneous HPC nodes.
- Developed distributed training pipelines using PyTorch and Flower across 40 NVIDIA GTX 1080Ti GPUs emulating 100 distributed sites — profiling throughput, communication overhead, and power utilization bottlenecks across the distributed training stack.

Sustainable Computing Lab, University of Massachusetts Amherst

Graduate Research Assistant

Amherst, MA

Sep 2021 – Present

PowerTrip — Distributed LLM Training Infrastructure

2024 – 2025

- Designed PowerTrip's dynamic greedy site selection algorithm — optimizing distributed training across GPU clusters by balancing aggregate compute power against WAN communication overhead, reducing time-to-accuracy by 50% vs. baselines (ACM SoCC '25).
- Engineered data-parallel distributed training infrastructure using Flower — implementing synchronous gradient aggregation, WAN-aware collective communication scheduling, and LoRA/QLoRA support for LLM training across geo-distributed GPU nodes.
- Built performance profiling tooling to identify bottlenecks across the training stack — analyzing power utilization, gradient synchronization latency, network throughput, and computation-communication overlap.
- Designed scheduling policies that adapt to dynamic site power availability and network conditions — maintaining consistent training progress under heterogeneous and time-varying distributed environments.

PowerScale — Hierarchical Aggregation Framework to Improve Training Efficiency

2025 – 2026

- Designed PowerScale, a hierarchical aggregation framework for distributed LLM and foundation model training — organizing GPU clusters into regional groups with synchronous intra-cluster and asynchronous inter-cluster communication, directly addressing computation-communication overlap and straggler mitigation at scale.
- Analyzed trade-offs between intra-cluster synchronization frequency, inter-cluster WAN communication cost, and cluster topology — characterizing how each dimension affects energy efficiency and convergence across heterogeneous GPU deployments.
- Developed K-means and agglomerative dynamic clustering for online GPU cluster topology adaptation at runtime based on real-time power and network conditions.
- Designed magnitude-based adaptive gradient synchronization — automatically reducing global communication frequency as training converges, improving throughput and energy efficiency.
- PowerScale reduces energy by 3.9× relative to single-tier FedAvg and 2.4× relative to PowerTrip — cutting communication energy by 10× and idle energy by 15× — while matching or slightly improving time-to-accuracy.

EcoLearn — Carbon Cost Optimization for Federated Learning

2023 – 2025

- Developed EcoLearn, an end-to-end federated learning training system that schedules distributed ML workloads to minimize the carbon emissions associated with large-scale FL training (ACM/IEEE SEC '25).

- Designed scheduling policy based on client utility and carbon intensities — selecting clients based on critical learning period while avoiding stragglers, reducing ML training carbon footprint by 10.8× while maintaining accuracy within ~1% of SOTA baselines.

Federated DGMR / FED-CAST — Generative Precipitation Nowcasting

2024 – 2026

- Deployed a federated generative deep learning framework (DGMR) for spatiotemporal precipitation nowcasting across geographically distributed U.S. climate regions using the Flower framework — building end-to-end distributed training pipelines for large-scale generative model training (AMS '25).
- Designed a federated training and evaluation workflow (FED-CAST) for generative precipitation nowcasting across 7 climate-diverse regional data sources — achieving up to 57.5% higher performance (TOPSIS) than centralized training in low-data regimes, while maintaining centralized accuracy at full data scale (IEEE eScience '26).

Cost Incentives for Users in Shared Cloud Clusters

2022 – 2023

- Analyzed an 8-year production cloud job trace (67M jobs, 1,790 users) to characterize job workload patterns, scheduling dynamics, and resource utilization across large-scale shared cloud clusters.
- Demonstrated that more than 80% of shared cluster users are financially incentivized to defect under different cloud pricing models (ICPE '23, Best Paper Runner-Up).

PUBLICATIONS

- [1] T. Mehboob, M. Zink, D. Irwin. "PowerScale: Improving Distributed Model Training Efficiency using Federated Datacenter Power." Under Submission.
- [2] T. Mehboob, L. Guo, N. Tallent, M. Zink, D. Irwin. "PowerTrip: Exploiting Federated Heterogeneous Datacenter Power for Distributed ML Training." ACM Symposium on Cloud Computing (SoCC '25).
- [3] T. Mehboob, N. Bashir, J. Iglesias, M. Zink, D. Irwin. "EcoLearn: Optimizing the Carbon Footprint of Federated Learning." ACM/IEEE Symposium on Edge Computing (SEC '25).
- [4] Z. Xu, T. Mehboob, A. Rutkovskii, M. Zink. "Leveraging Federated Learning for Generative Precipitation Nowcasting in Diverse U.S. Climate Regions." AMS Annual Meeting (AMS '25).
- [5] T. Mehboob, N. Bashir, D. Irwin, M. Zink. "Is Sharing Caring? Analyzing the Incentives for Shared Cloud Clusters." ACM/SPEC International Conference on Performance Engineering (ICPE '23). Best Paper Runner-Up.
- [6] Z. Xu, T. Mehboob, M. Zink, C. Davis. "FED-CAST: Federated Training and Evaluation Workflow for Generative Precipitation Nowcasting Across Climate-Diverse Regions." IEEE International Conference on e-Science (eScience '26).

TECHNICAL SKILLS

Distributed Training	Large-Scale GPU Cluster Training, Data Parallelism, Gradient Synchronization, WAN-Aware Scheduling, Computation-Communication Overlap, Fault-Tolerant Training, Straggler Mitigation, Hierarchical Aggregation
LLM Infrastructure	vLLM, LLM Training & Inference, Prefill/Decode Disaggregation, KV Cache Optimization (FP8/INT8), LoRA/QLoRA, Model Quantization (INT4/FP8), SLO-Aware Scheduling, Hugging Face
GPU & Systems	CUDA, NVML, NCCL, Collective Communication, GPU Profiling & Benchmarking, Power-Aware Scheduling, Throughput & Latency Optimization
Data & Analysis	Large-Scale Data Processing, Distributed Dataset Management, Pandas, NumPy, SQL
Observability & Ops	Training Monitoring & Telemetry, Performance Profiling, Bottleneck Analysis, Kubernetes, Docker, Slurm, AWS, Linux
Frameworks & Languages	PyTorch, Flower, Ray, TensorFlow, Python, C/C++, CUDA C, Bash/Shell, Git

AWARDS & HONORS

- Research Talk Winner — Engineering Research Symposium, UMass Amherst (2026).
- Best Paper Runner-Up — ACM/SPEC International Conference on Performance Engineering (ICPE '23).
- Rector's Gold Medal — Best Final Year Project & Thesis, entire graduating class at NUST.
- Visiting Researcher (Fully Funded) — Technical University of Munich (TUM).
- Academic Merit Scholarships — Awarded for sustained excellence throughout high school and undergraduate studies.